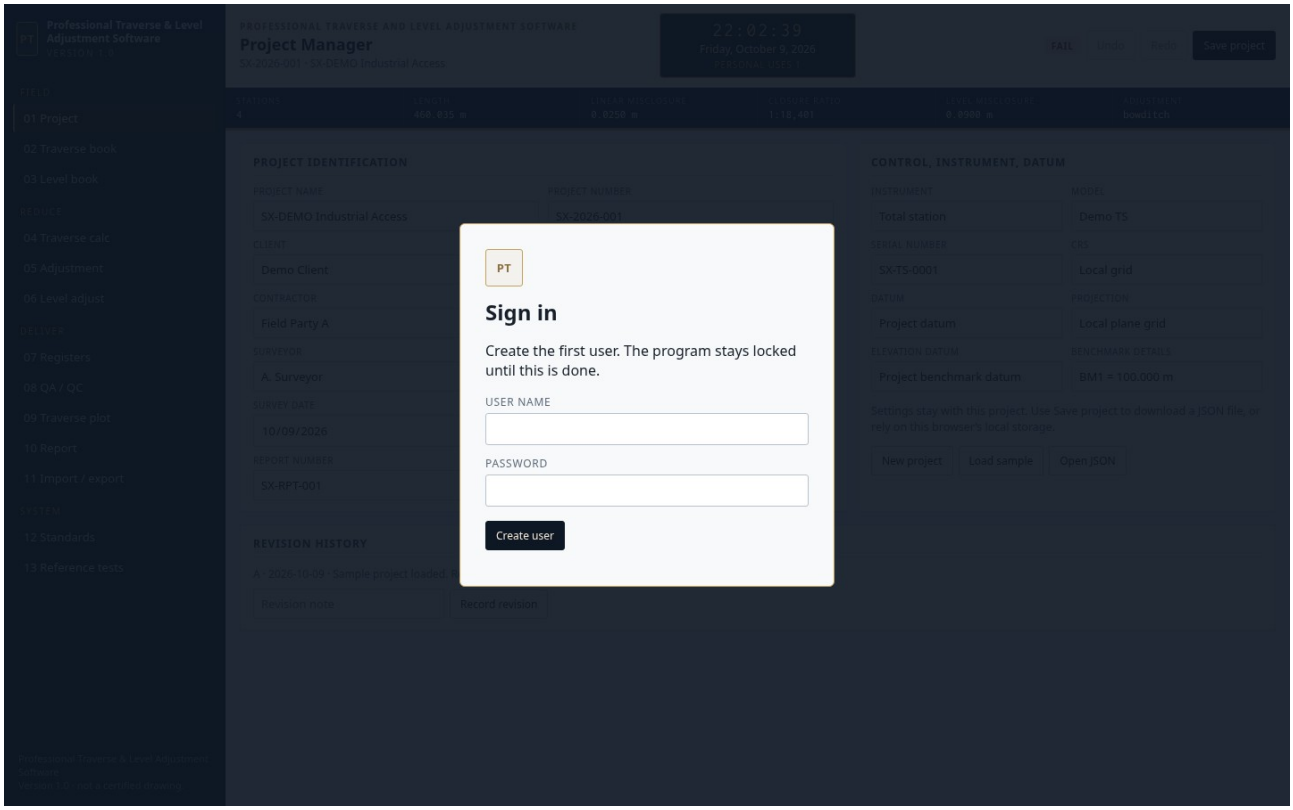
Professional Traverse & Level Adjustment Software

Complete user manual. Each step uses a screen from the program.

1. Install and sign in

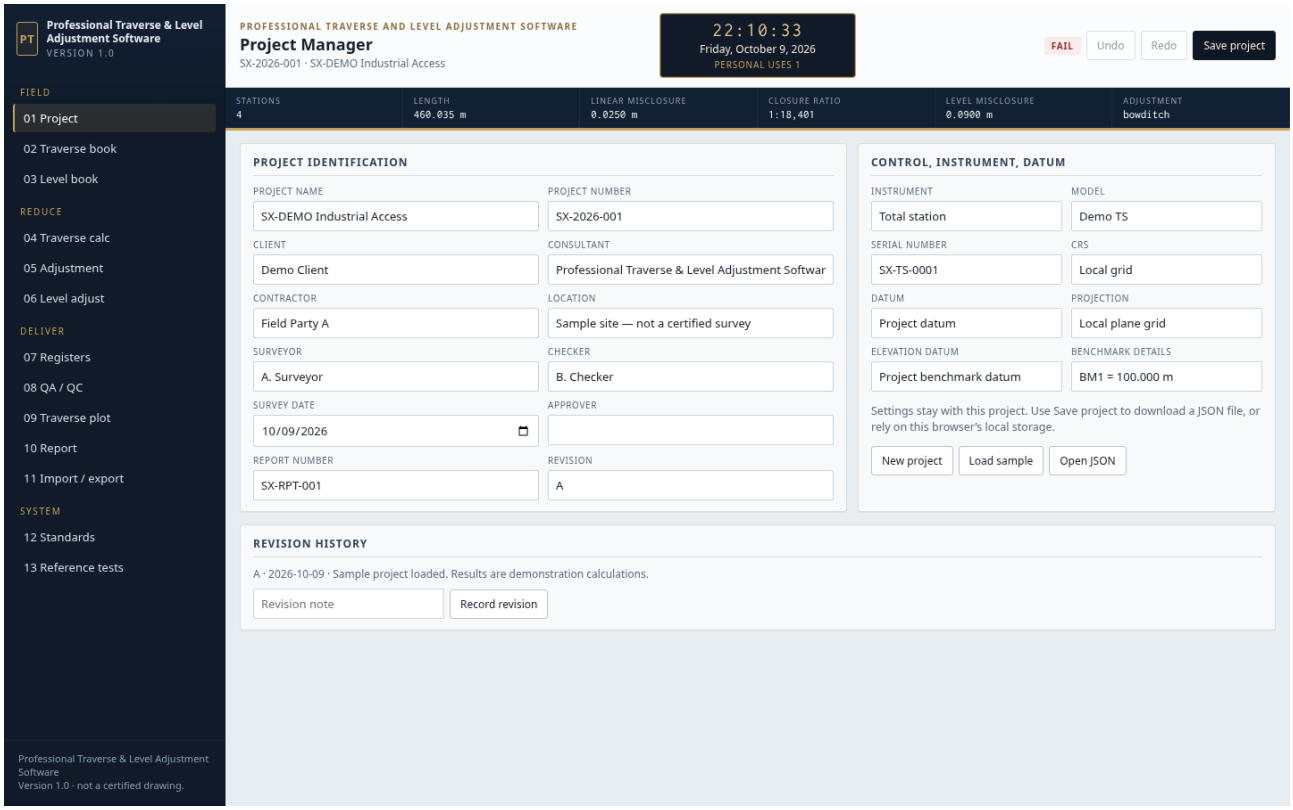
Run Professional-Traverse-Level-Adjustment-Setup-1.0.exe. Open the desktop shortcut. The program stays locked until a user is created. Enter a user name and password, then choose Create user. Next time, the same name and password are required. In 12 Standards you can add another user or sign out.



Screen from the program.

2. Project Manager

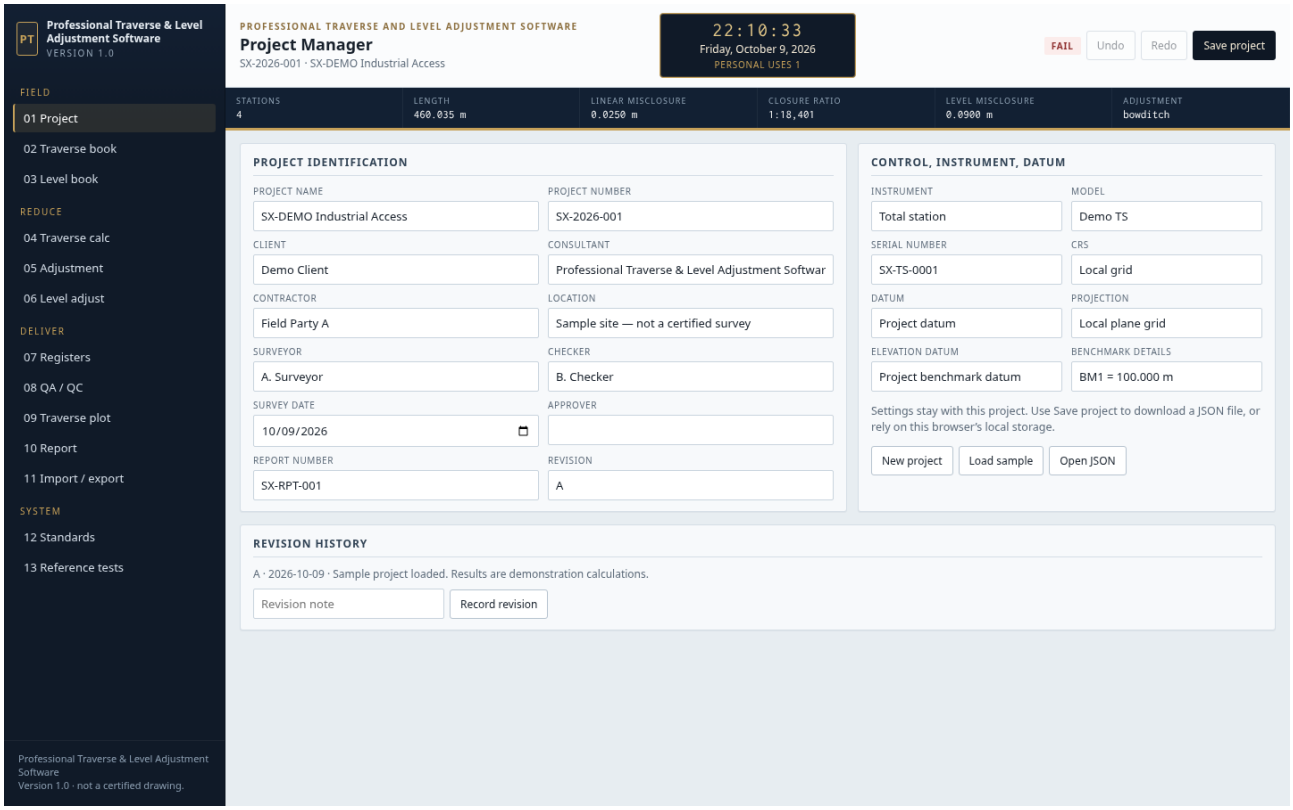
The first screen after sign-in is Project Manager. Enter the project name, number, client, surveyor, checker, instrument, datum and benchmark. New project clears the job. Load sample fills the demonstration traverse and level run. Open JSON reads a saved project. Save project downloads the job as a JSON file.



Screen from the program.

3. Clock, calendar and personal use

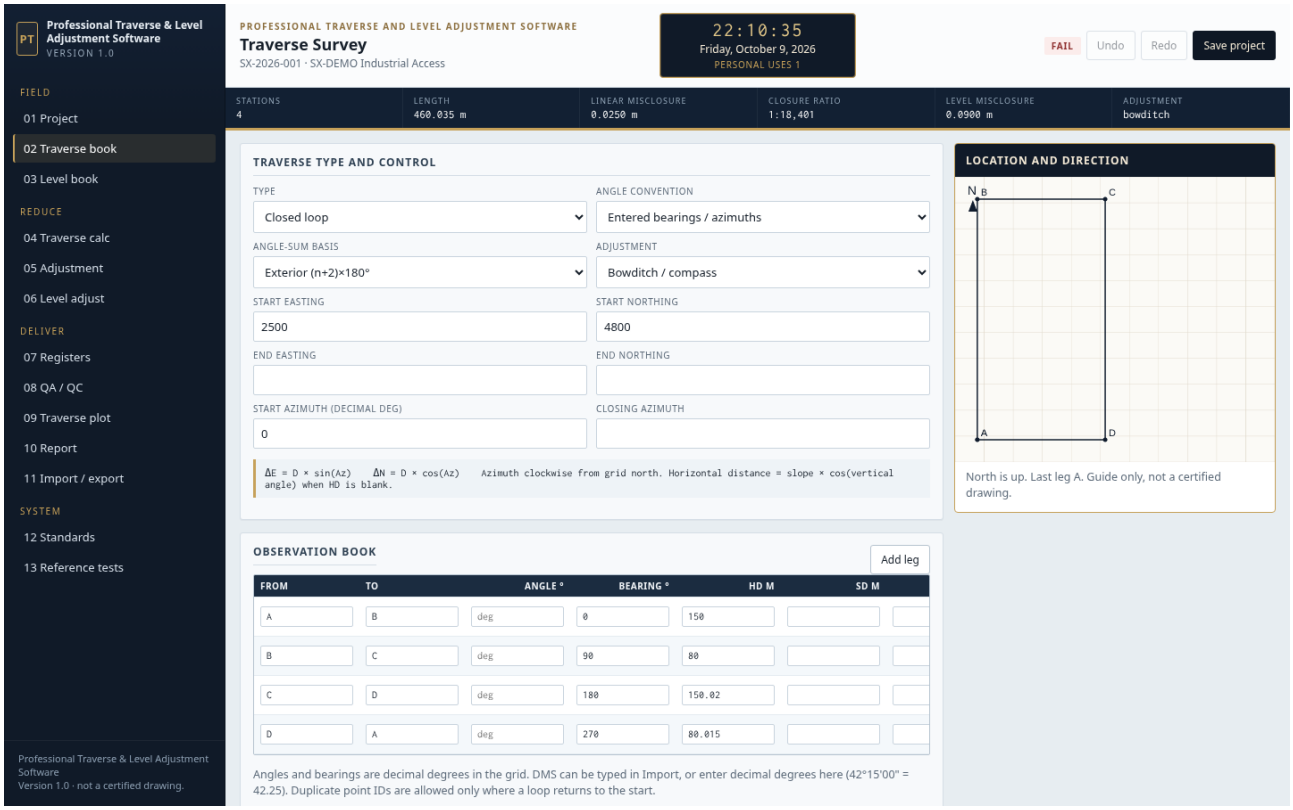
The gold clock and calendar stay at the top centre on every screen. Personal uses increases once each time this computer opens the program. The count is stored only on this computer.



Screen from the program.

4. Traverse book and live plan

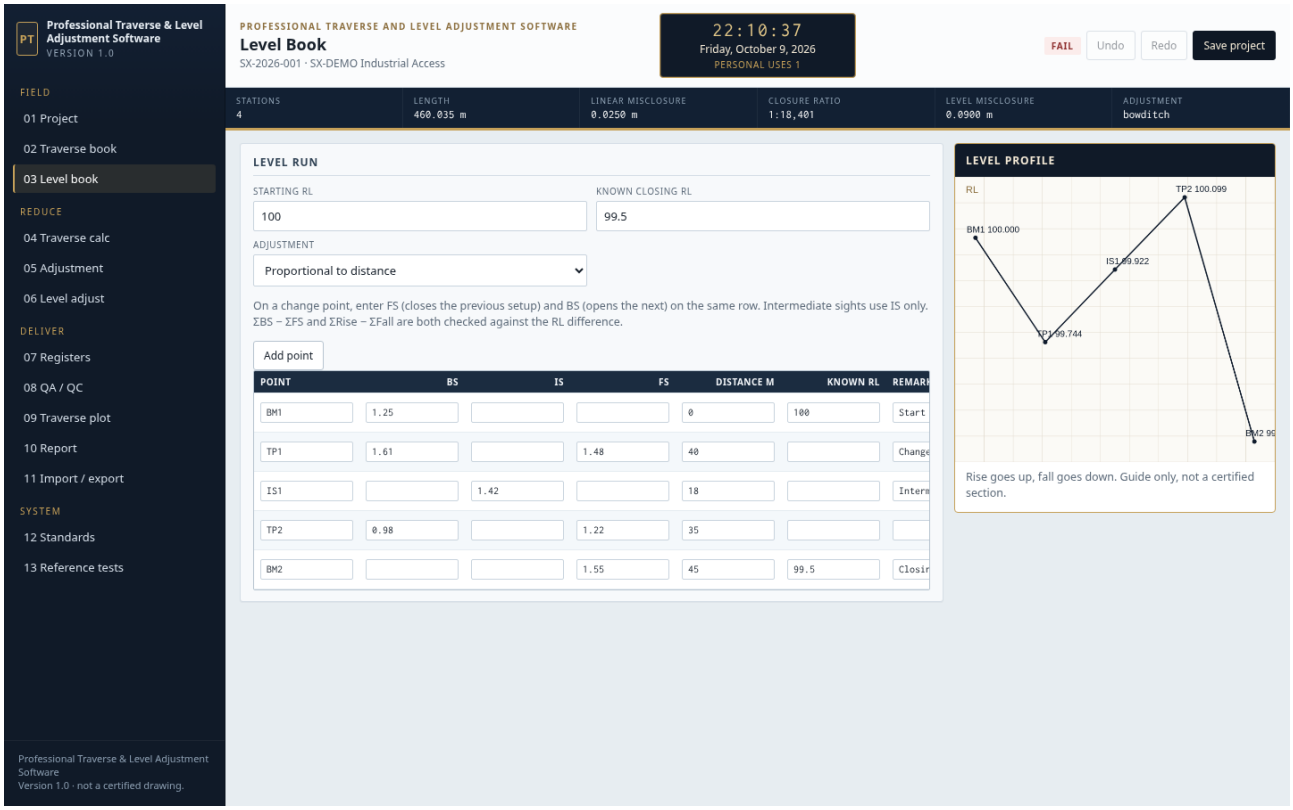
Open 02 Traverse book. Choose the traverse type, angle convention and adjustment method. Enter the start easting and northing, and the end control if the traverse closes on a known point. Add each leg with from, to, angle or bearing, and horizontal distance. The Location and Direction window on the right draws the stations as you type. North is up. This plan is a guide, not a certified drawing.



Screen from the program.

5. Level book and profile

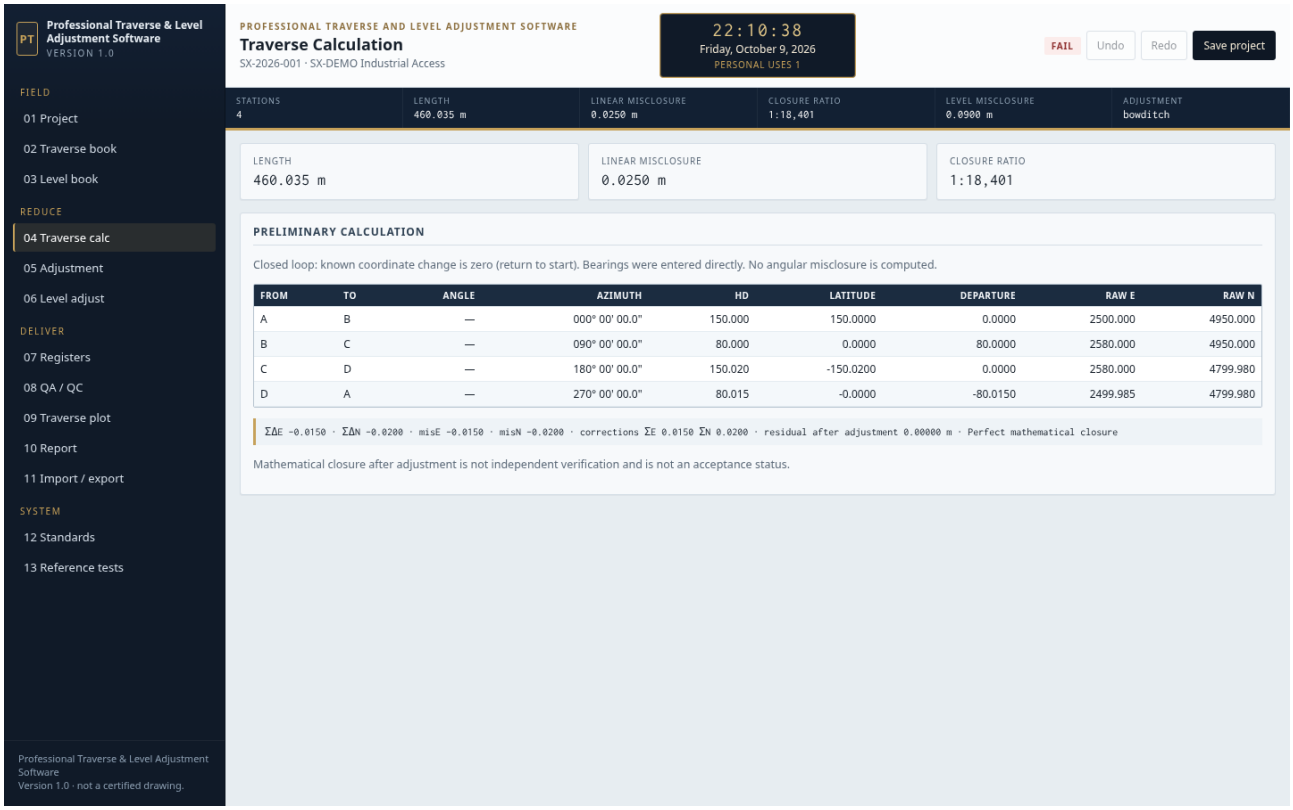
Open 03 Level book. Enter the starting reduced level and, if the run closes, the known closing reduced level. On a change point enter the foresight and the next backsight on the same row. Intermediate sights use IS only. The Level Profile window shows the rise and fall. A rise goes up and a fall goes down.



Screen from the program.

6. Traverse calculation

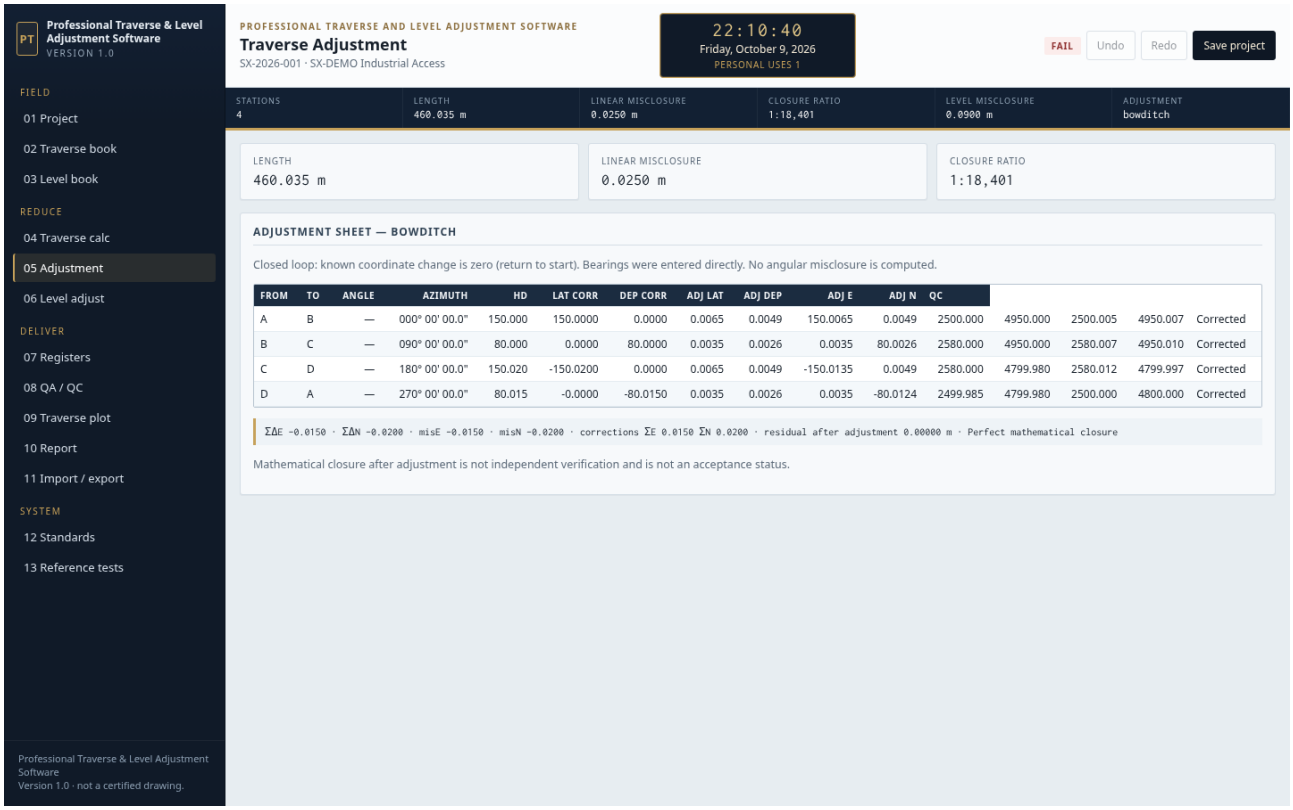
Open 04 Traverse calc. The sheet shows each leg, azimuth, latitude, departure and raw coordinates. Azimuth is clockwise from grid north. Easting change is distance times sine of azimuth. Northing change is distance times cosine of azimuth. An open traverse with no closing control does not claim a closing error.



Screen from the program.

7. Traverse adjustment

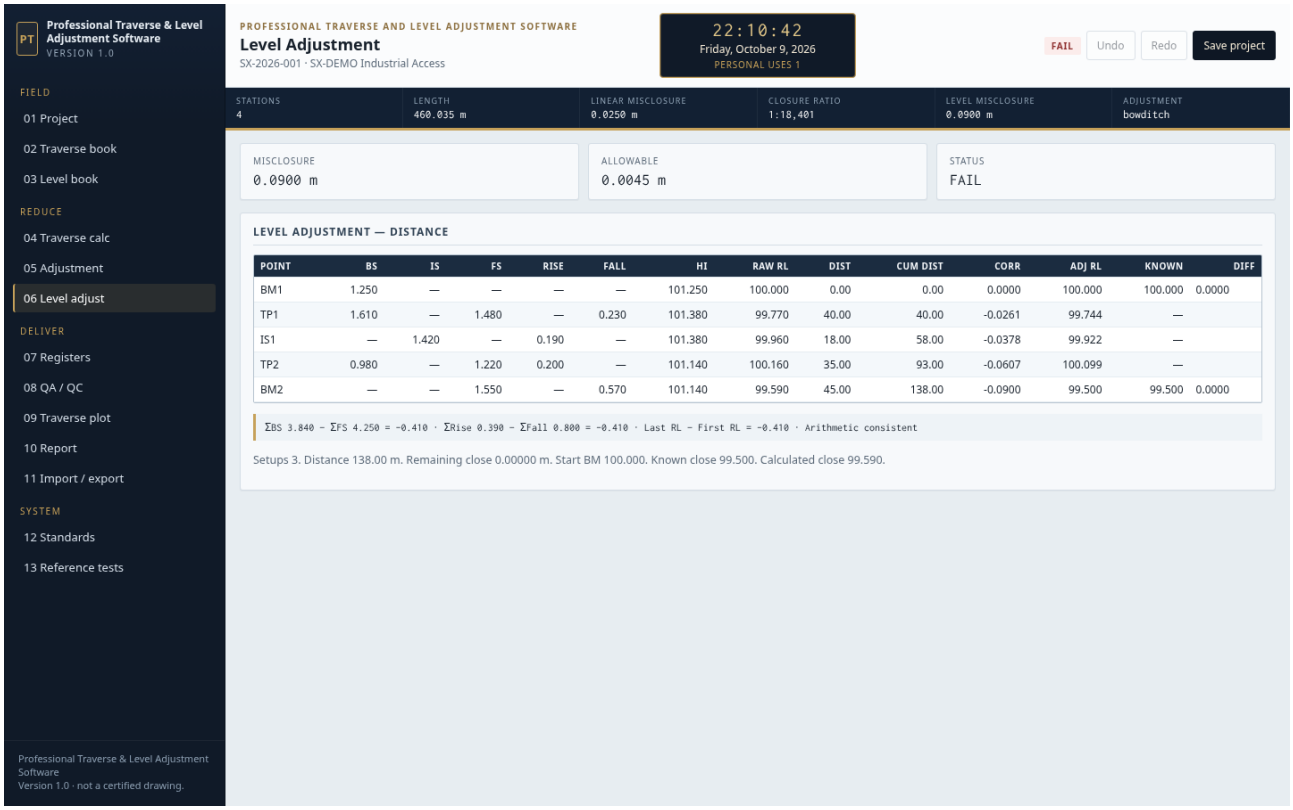
Open 05 Adjustment. Choose Bowditch, Transit, weighted least squares, or no adjustment. The sheet shows the misclosure, the closure ratio, and the correction on each leg. Final adjusted coordinates are separate from the raw coordinates. A mathematical close is not an acceptance decision.



Screen from the program.

8. Level adjustment

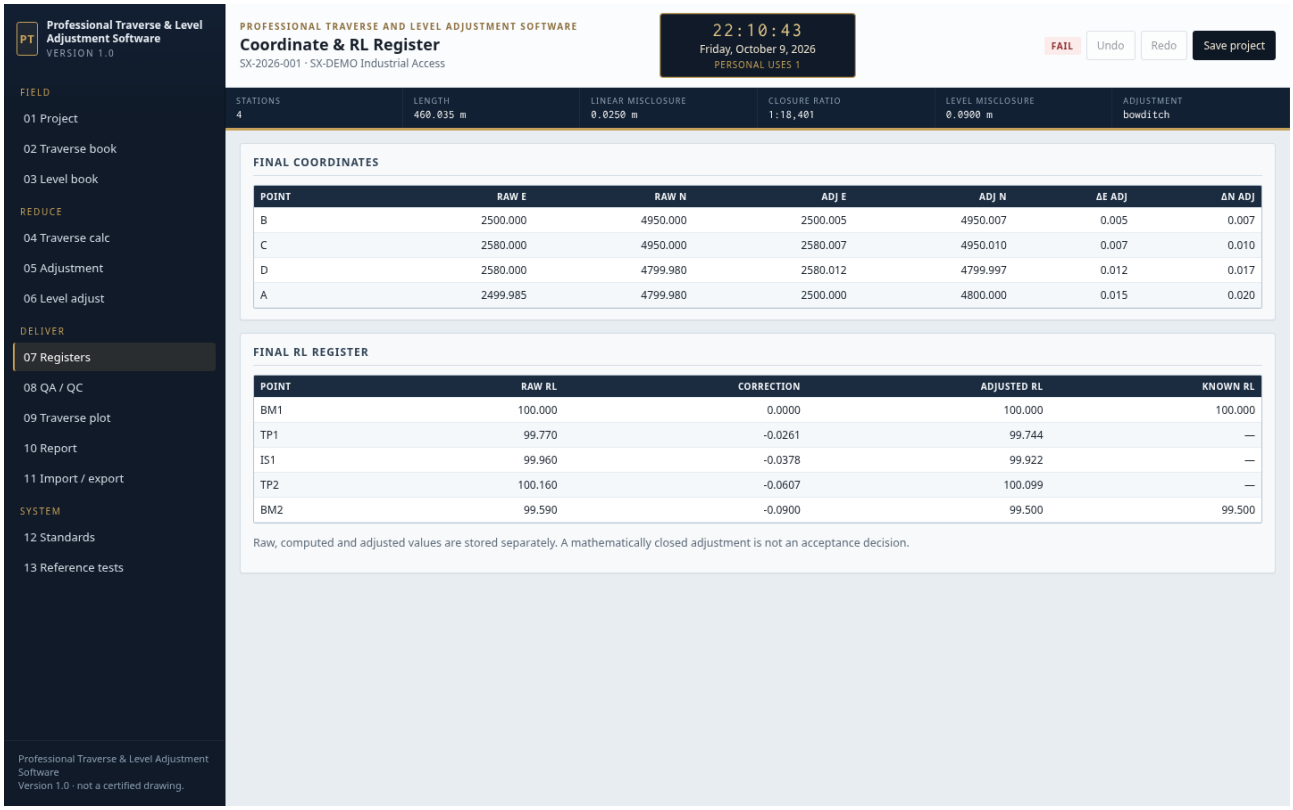
Open 06 Level adjust. The sheet checks the sum of backsights minus the sum of foresights against the reduced-level difference. It reports the misclosure and the allowable tolerance from Settings. Corrections and adjusted reduced levels are shown for each point.



Screen from the program.

9. Coordinate and RL register

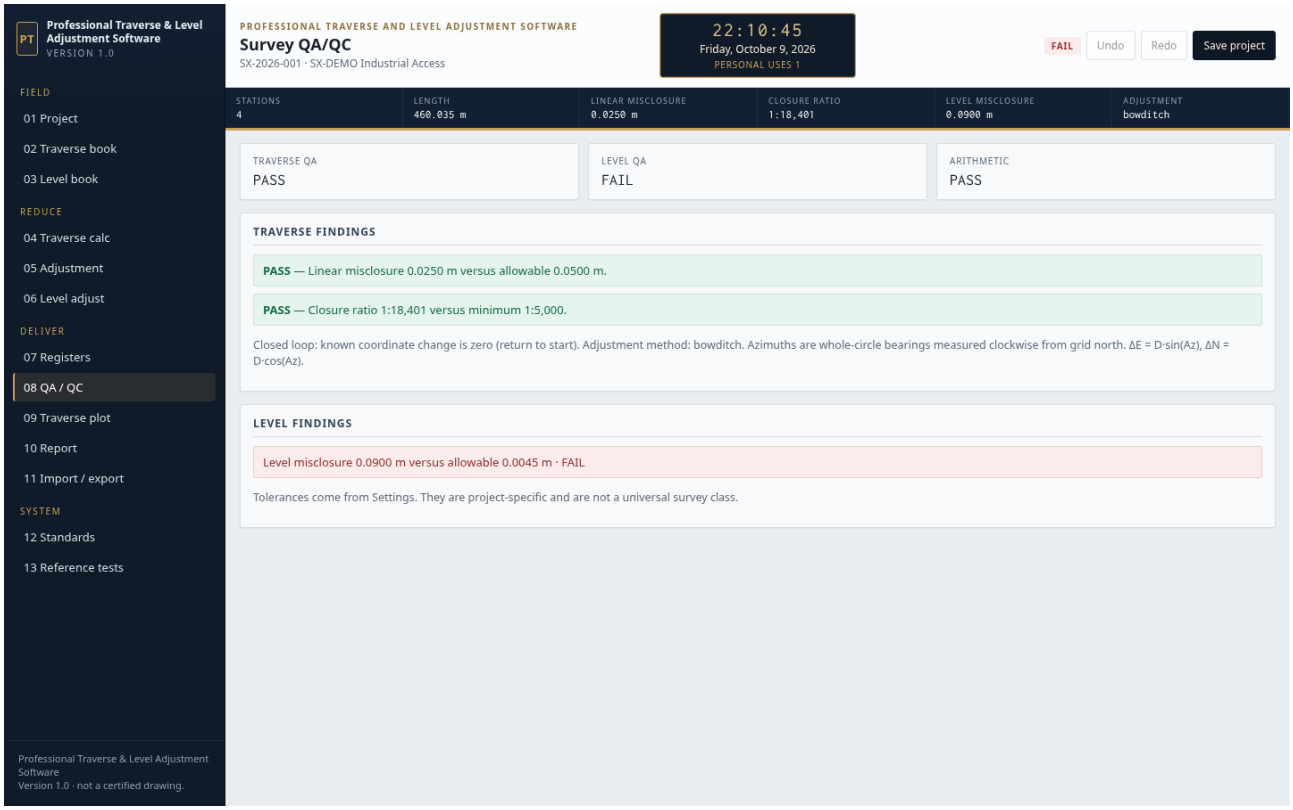
Open 07 Registers. This is the list of raw and adjusted easting and northing, and the raw and adjusted reduced levels. Use this sheet when the coordinates are copied to a drawing or a report.



Screen from the program.

10. Survey QA / QC

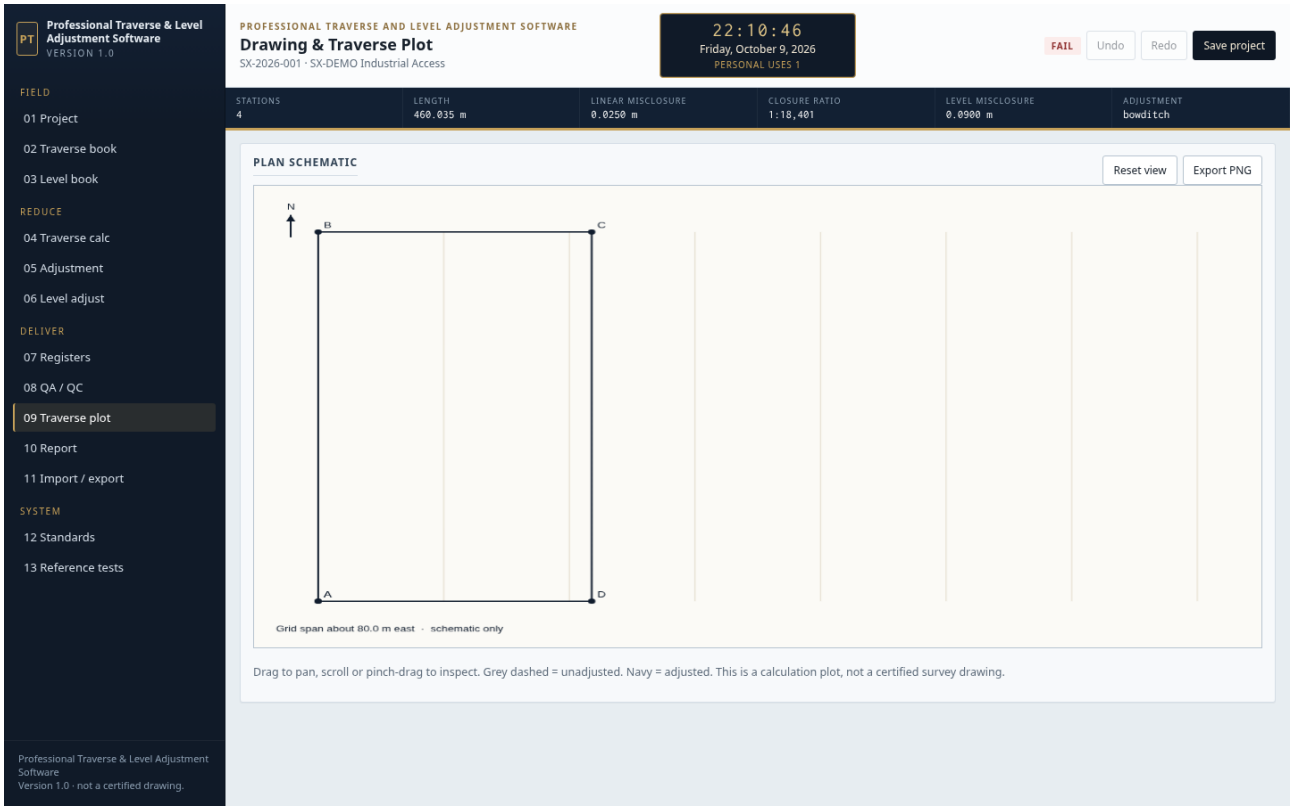
Open 08 QA / QC. The screen shows pass, warning or fail for the traverse and the level run. Tolerances come from 12 Standards. They are project limits, not a universal survey class.



Screen from the program.

11. Traverse plot

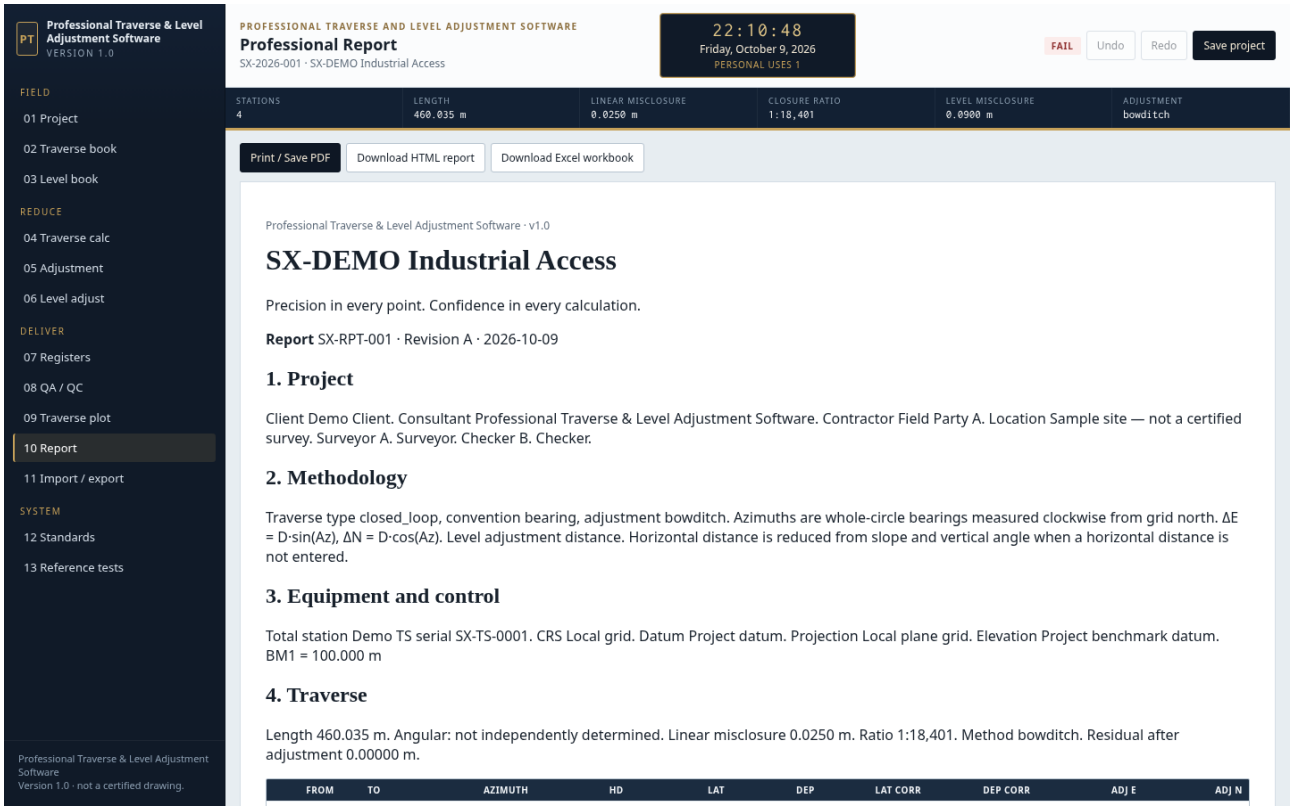
Open 09 Traverse plot. The full plan shows the adjusted line, the unadjusted line, station names and north. Drag to move the view and use the wheel to zoom. The plot is schematic only.



Screen from the program.

12. Professional report

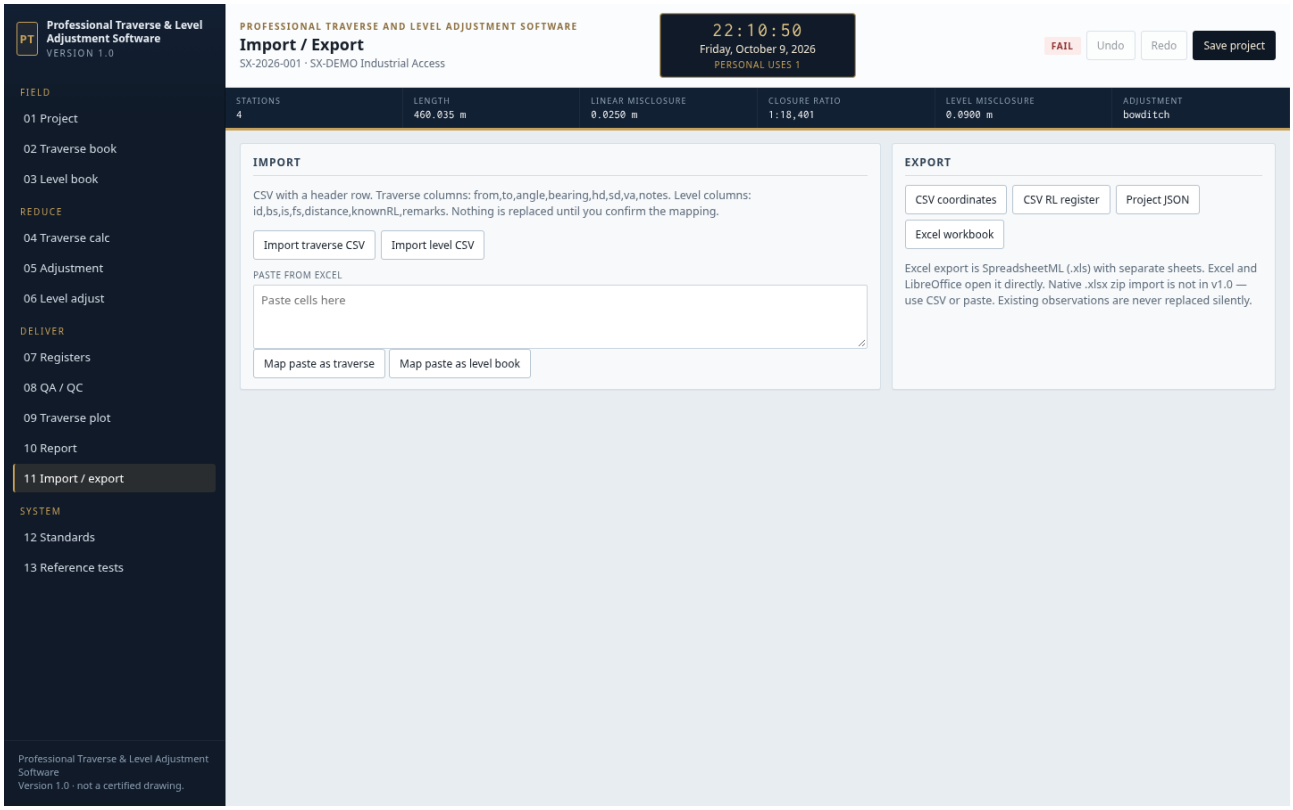
Open 10 Report. Review the project, traverse and level results, then print or save the HTML report. The workbook export is on 11 Import / export.



Screen from the program.

13. Import and export

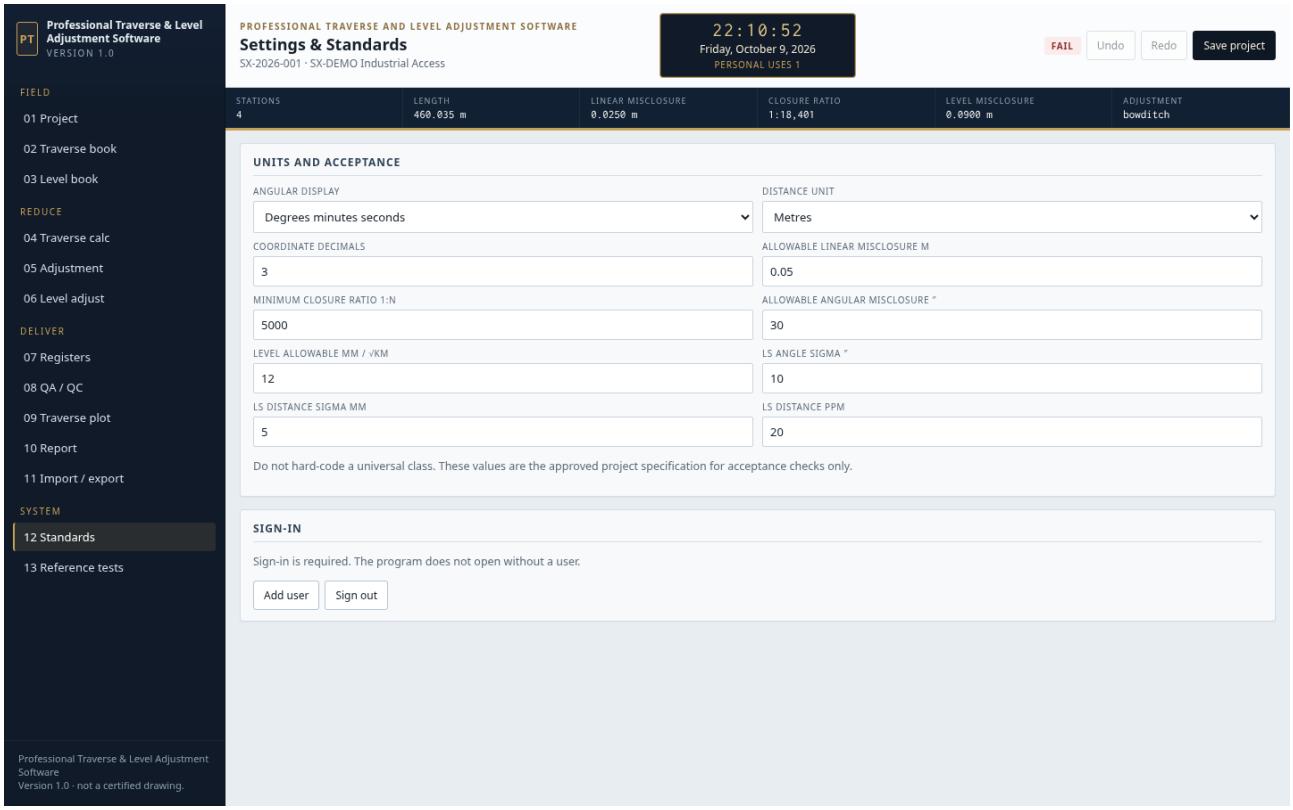
Open 11 Import / export. Paste or open a CSV for the traverse book or the level book, check the column map, then import. Export the workbook when the calculations are ready.



Screen from the program.

14. Standards

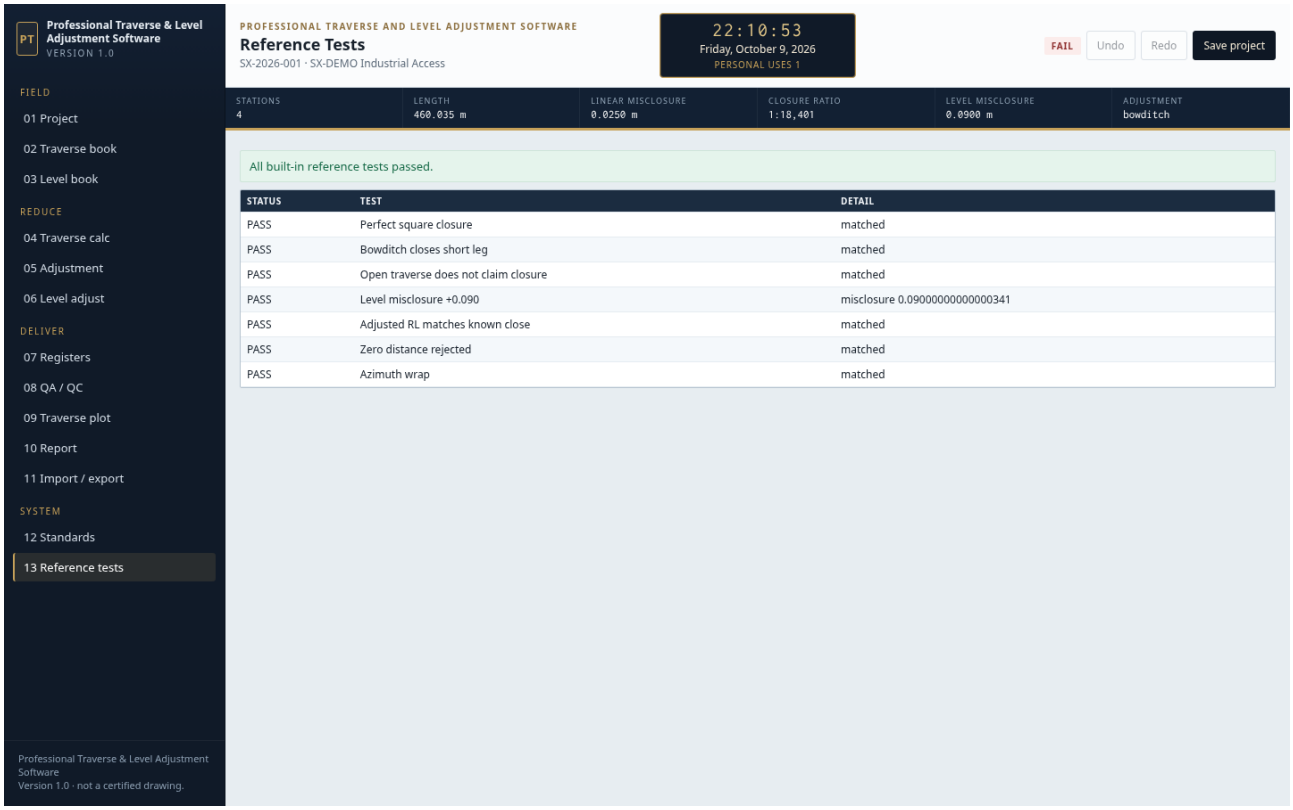
Open 12 Standards. Set the angle display, distance unit, coordinate decimals, allowable linear misclosure, minimum closure ratio, angular misclosure and level tolerance. Sign-in options are on this screen: add a user, or sign out.



Screen from the program.

15. Reference tests

Open 13 Reference tests. This runs the built-in checks for the traverse and level arithmetic. A pass here confirms the calculation engine, not the field work.



Screen from the program.